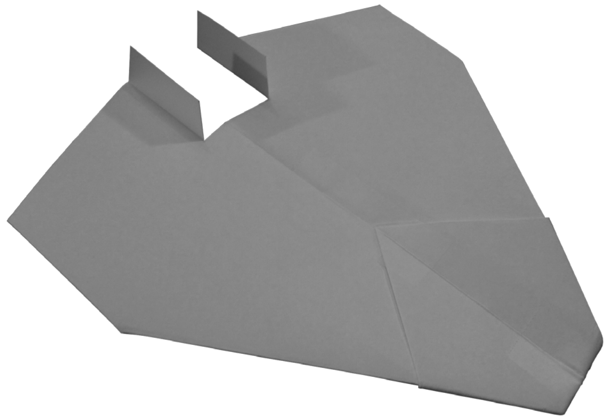
P16 NASP

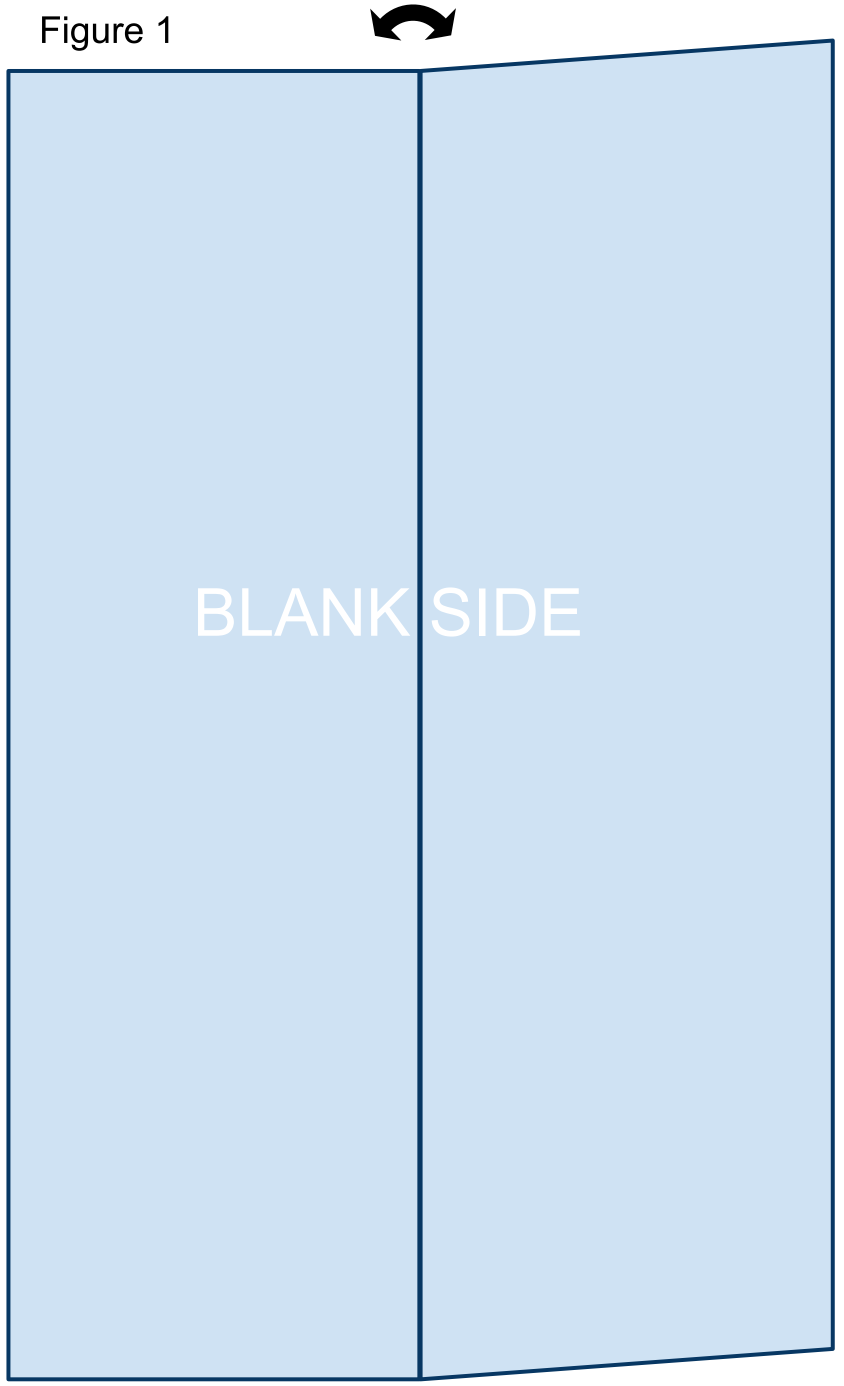


The printable template linked below is 4.25in x 5.5in. That is equivalent to half of an 8.5in x 11in sheet of printer paper. While the designs in this book will fly great with regular printer paper, I recommend 32lb/ream paper. The stronger paper will hold its shape better during flight allowing for higher and more exciting flights. Once you become comfortable making the planes from this template you can simply cut regular printer paper in half in order to start. Print the PDF template located at the bottom of the web page: https://sites.google.com/site/afspaperairplanes/p16

In case your reading device is not connected to the internet, doing a Google search for “AFS Paper Airplanes” should get you to the webpage.

1

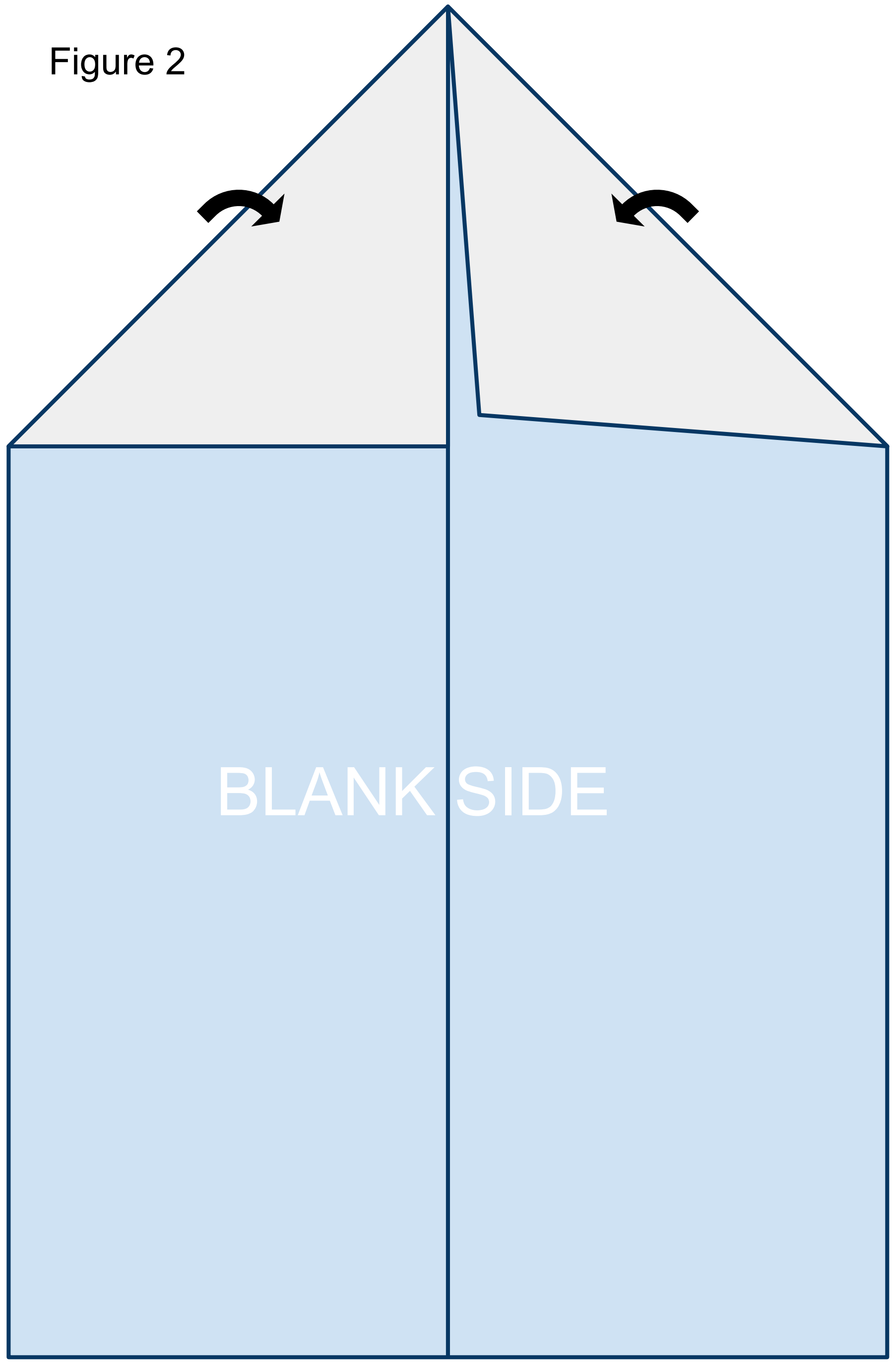
Cut out the template and place the paper on the table so that the printed information is on the underside with the printed triangle facing away from you. You will be looking at the blank side. Bring the right side up and fold the paper in half the long way. Unfold leaving a visible crease as shown in Figure 1 below.



2

Fold the leading corners back along the printed solid lines. The edge of the paper should line up along the crease you created in step 1 as shown in Figure 2 below.

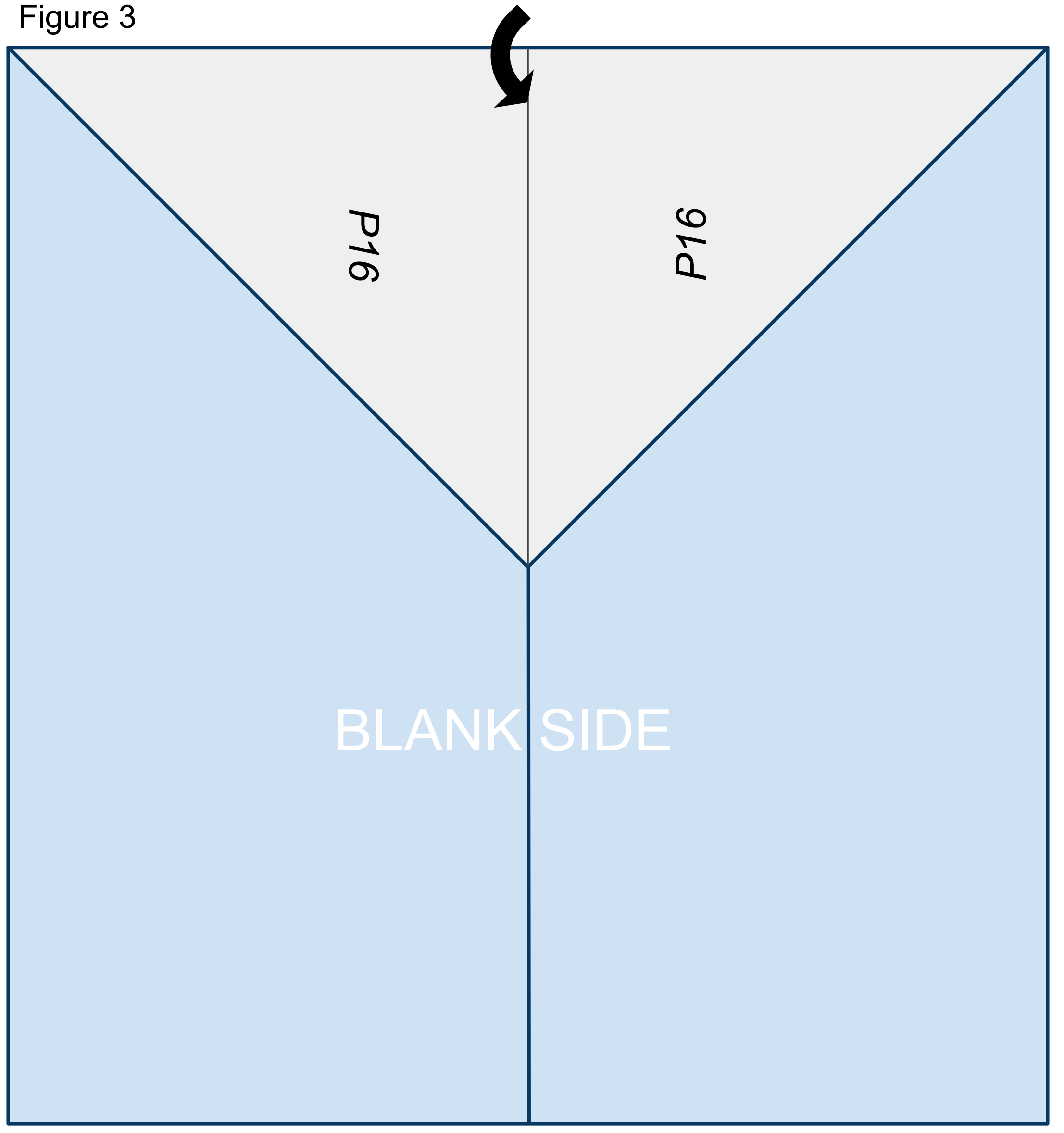
Figure 2



3

Fold the nose of the plane created in step 2 back along the solid line printed on the front of the paper so that it lines up on the center crease created in step 1 as shown in Figure 3 below.

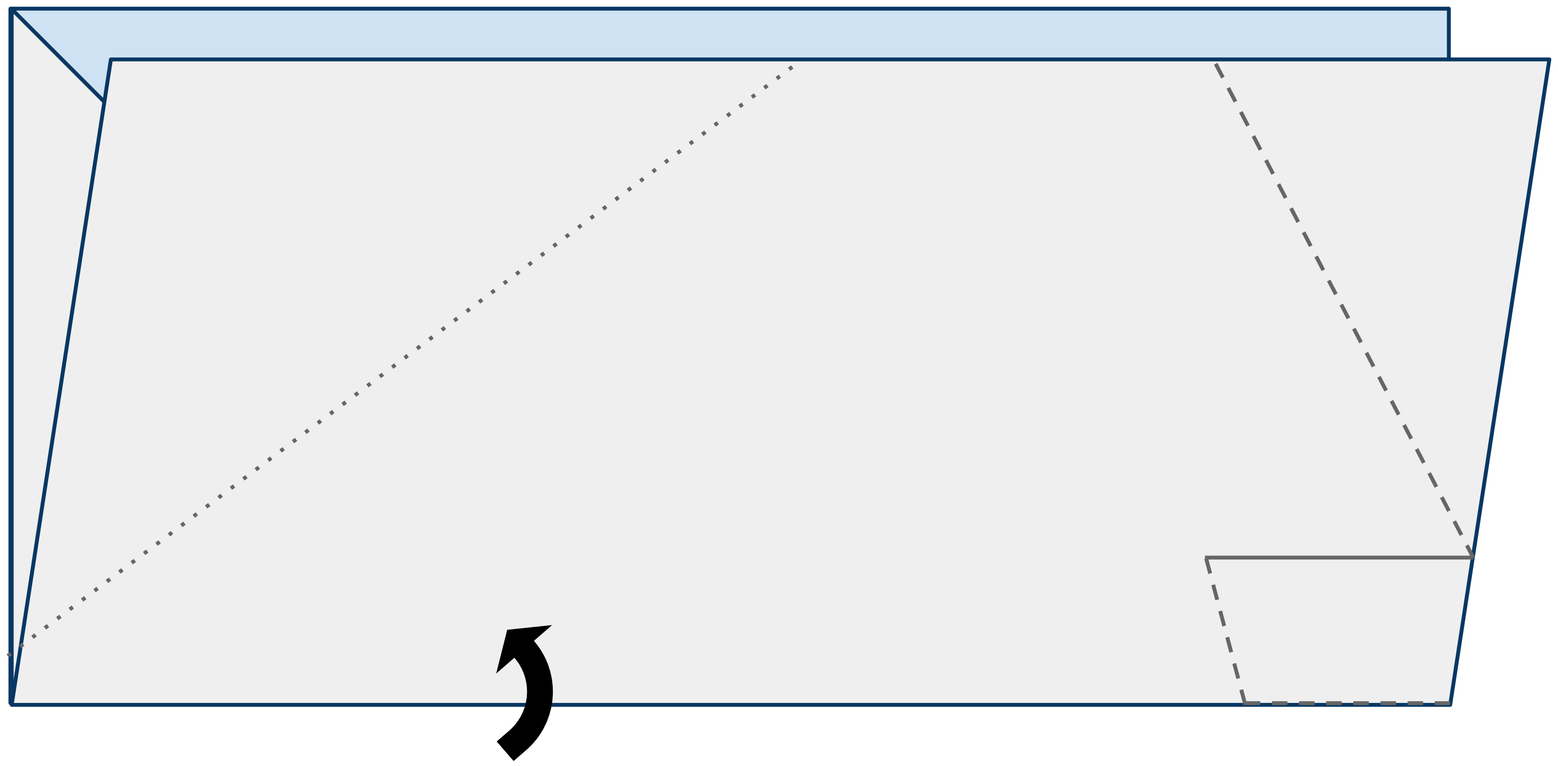
Figure 3



4

Turn the paper to the left (counterclockwise) and fold the paper upward in half hiding the triangle created in the previous steps refolding along the fold from step 1 as shown in Figure 4 below.

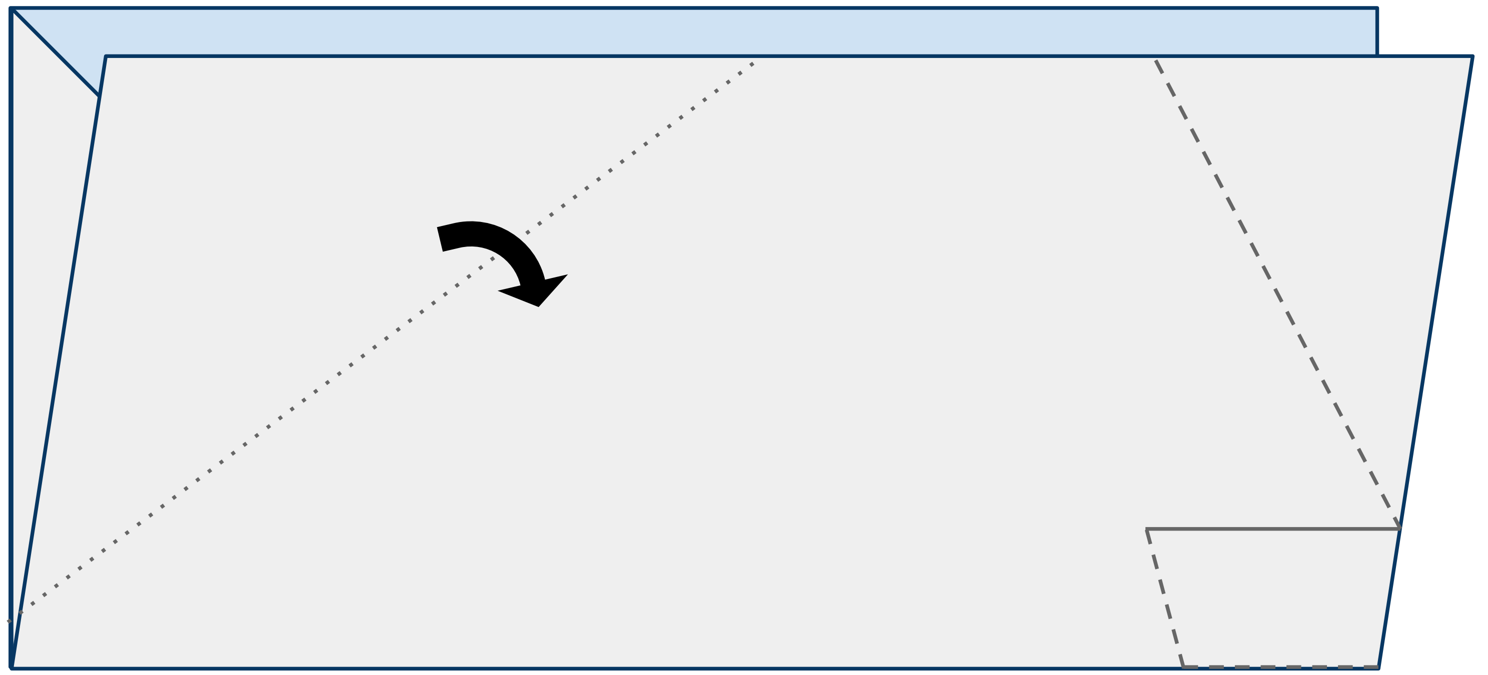
Figure 4



5

Fold the leading edge of the near side of the plane down along the dotted line as shown in Figure 5 below.

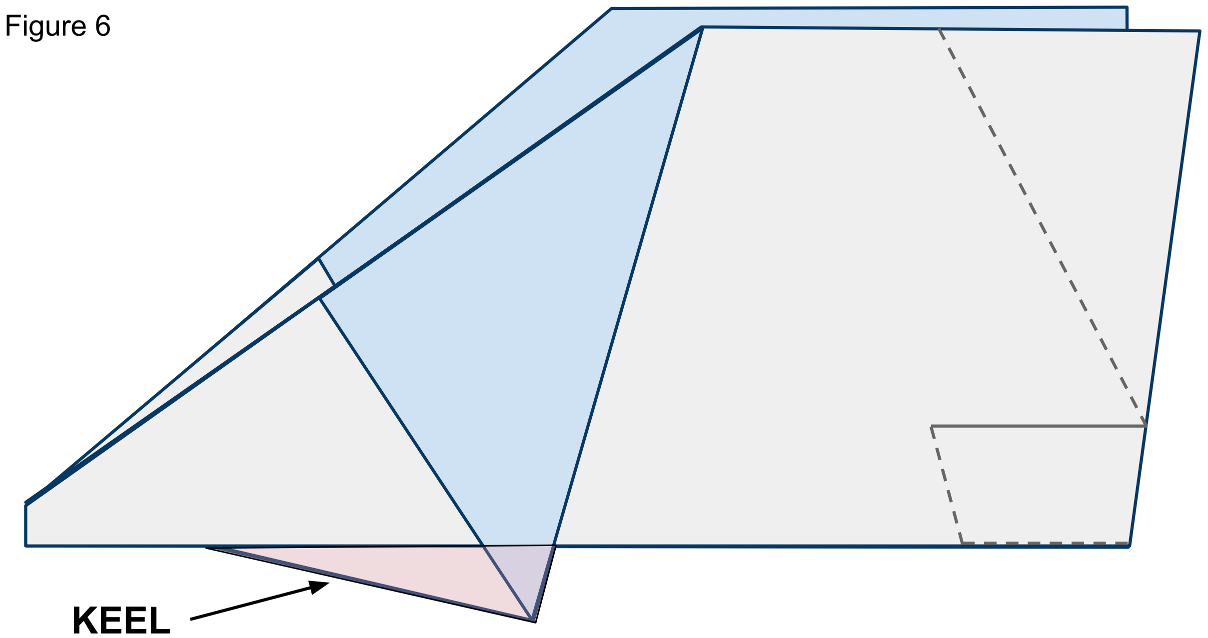
Figure 5



6

Do the same to the other side of the plane. It is important to match these 2 folds exactly so that your plane is symmetrical and will fly straight. The area that is created below the body and designated in pink will be called the keel.

Figure 6



7

Place tape along the keel as shown in Figure 7 below and cut a slit in the tape along the red line. Now fold the tape around and under both sides of the plane and keel securing them together. See Figure 7b for what the far side of the plane looks like.

Figure 7

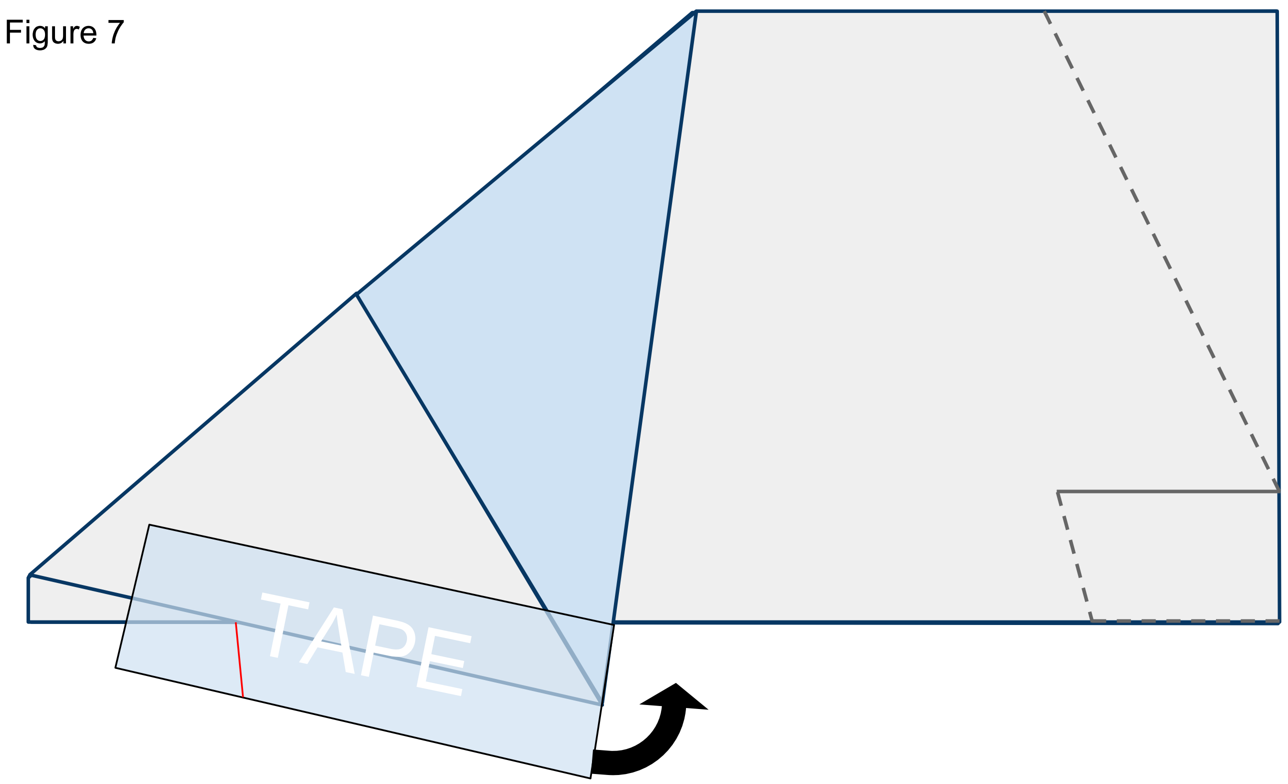
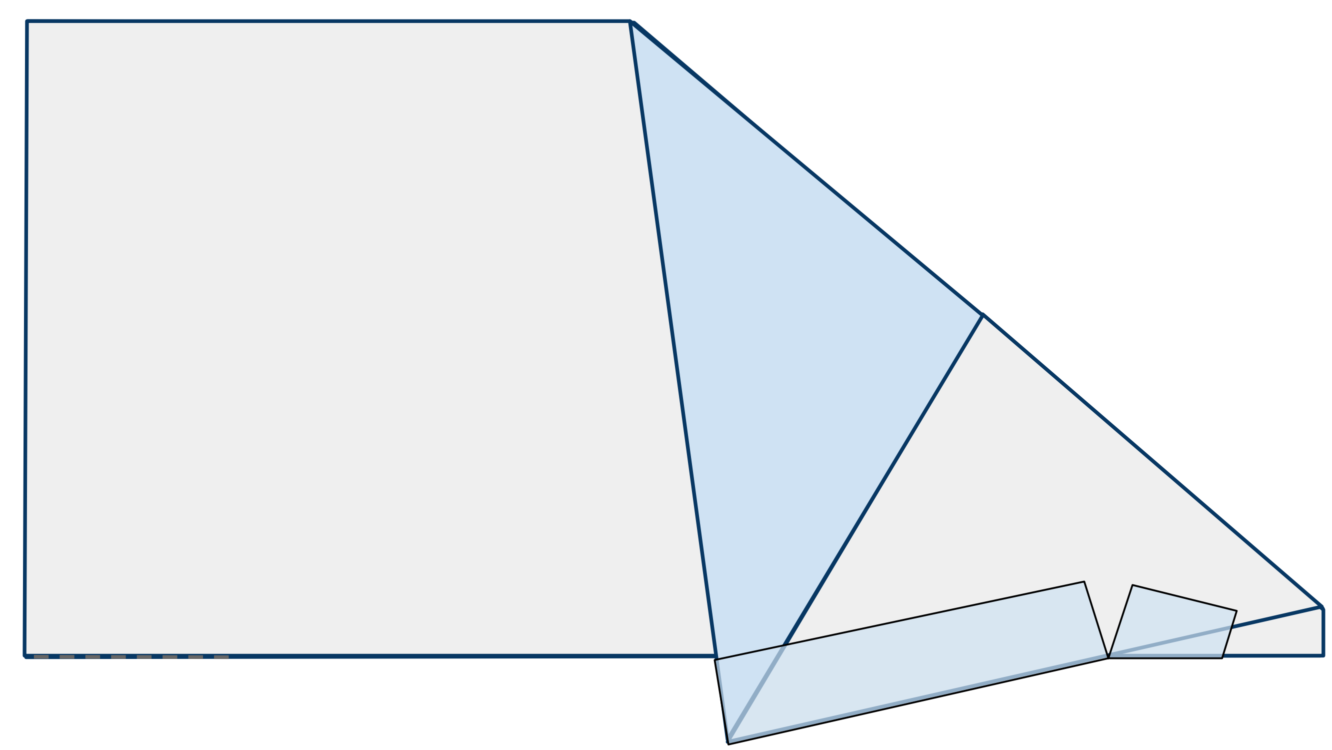


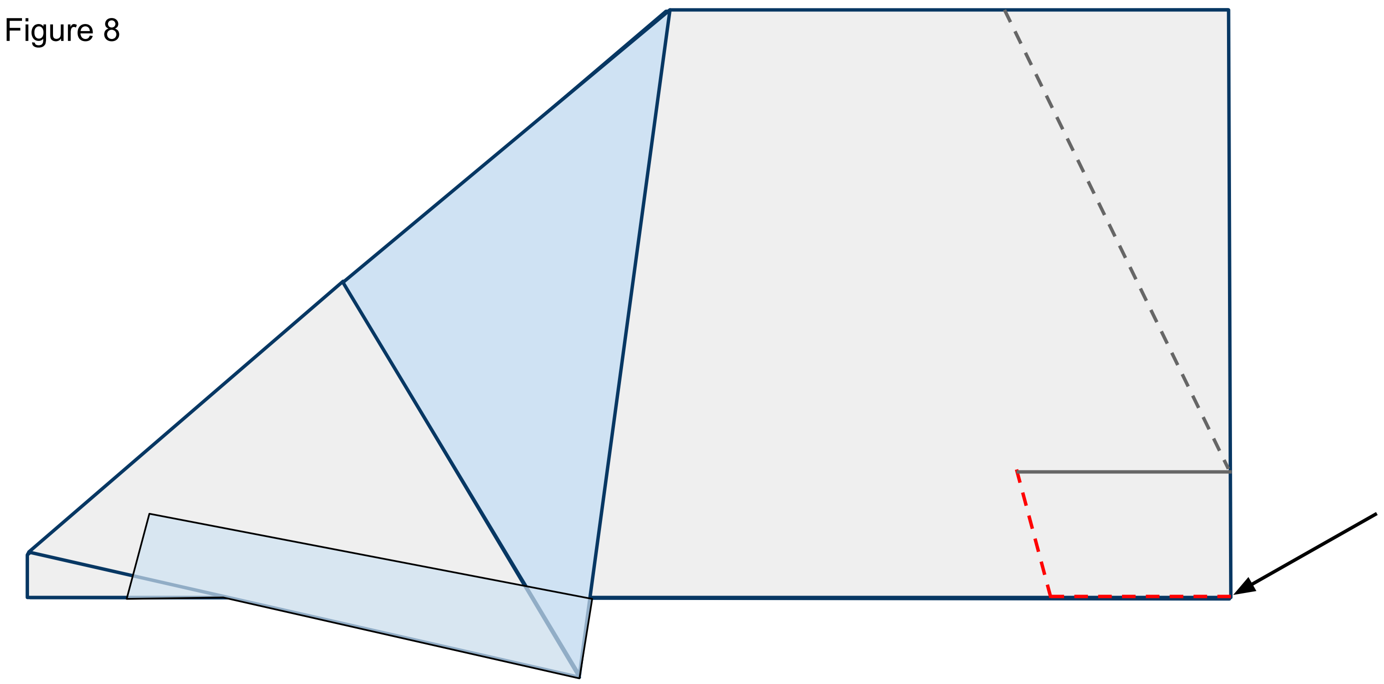
Figure 7b



8

Cut along the red dashed lines cutting through both sides of the plane as shown in Figure 8 below.

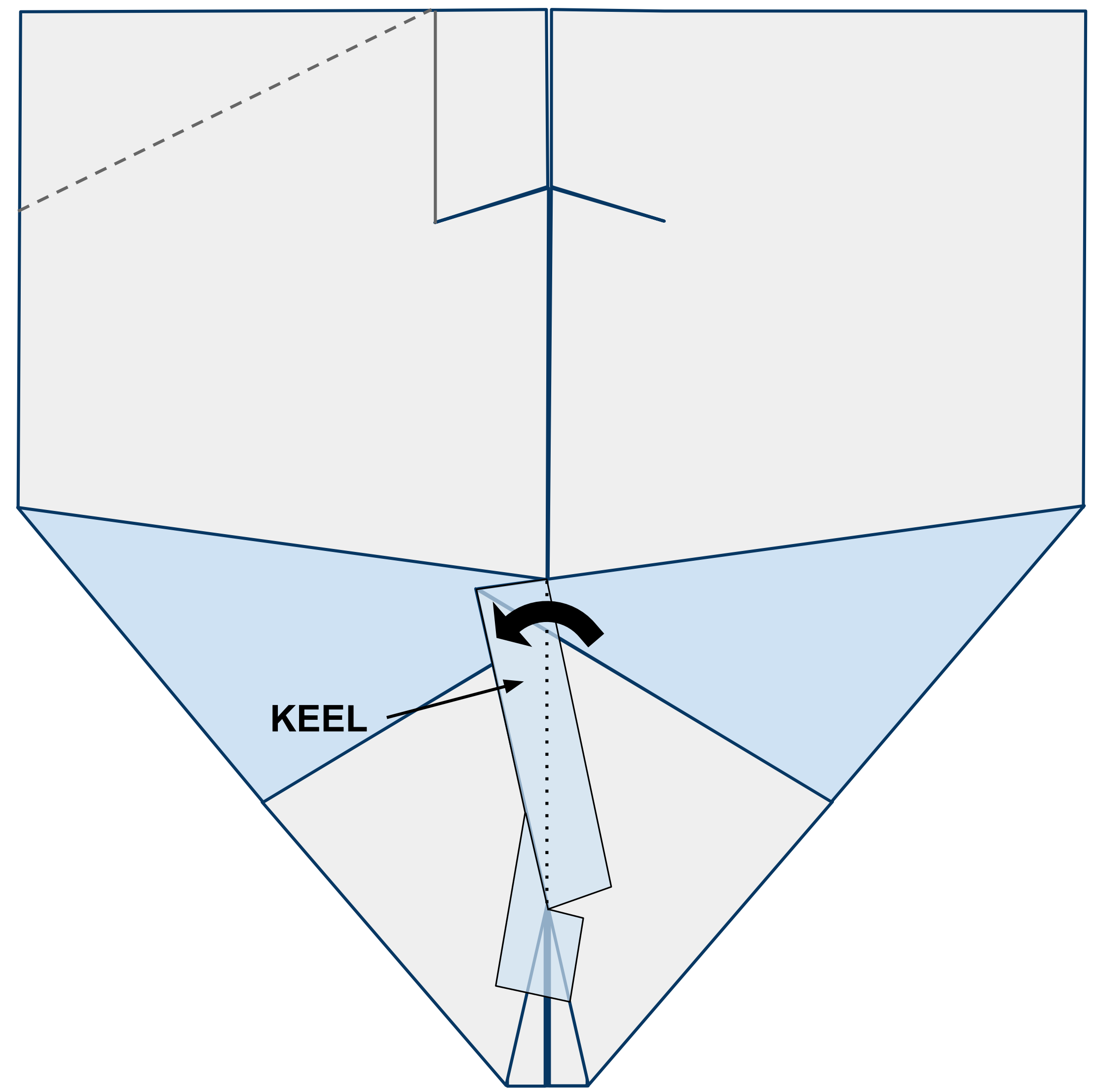
Figure 8



9

Open the plane up flat on your work surface with the nose toward you and looking at the underside of the body. When you do this the keel will want to stand up off the table. Bend it to the left along the center line and give it a strong crease as shown in Figure 9 below.

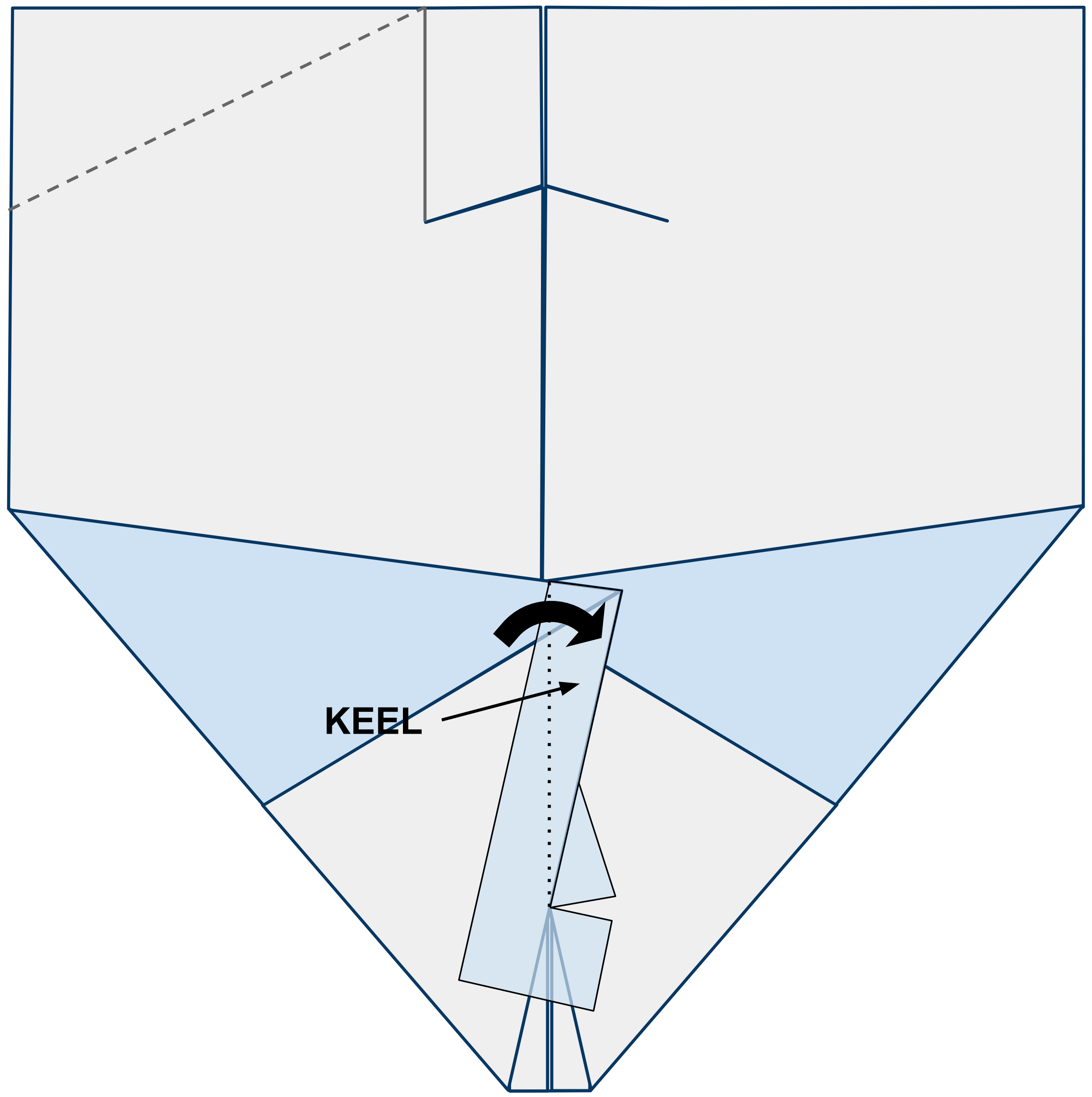
Figure 9



10

Now fold the keel to the right side creating a strong crease as shown in Figure 14 below.

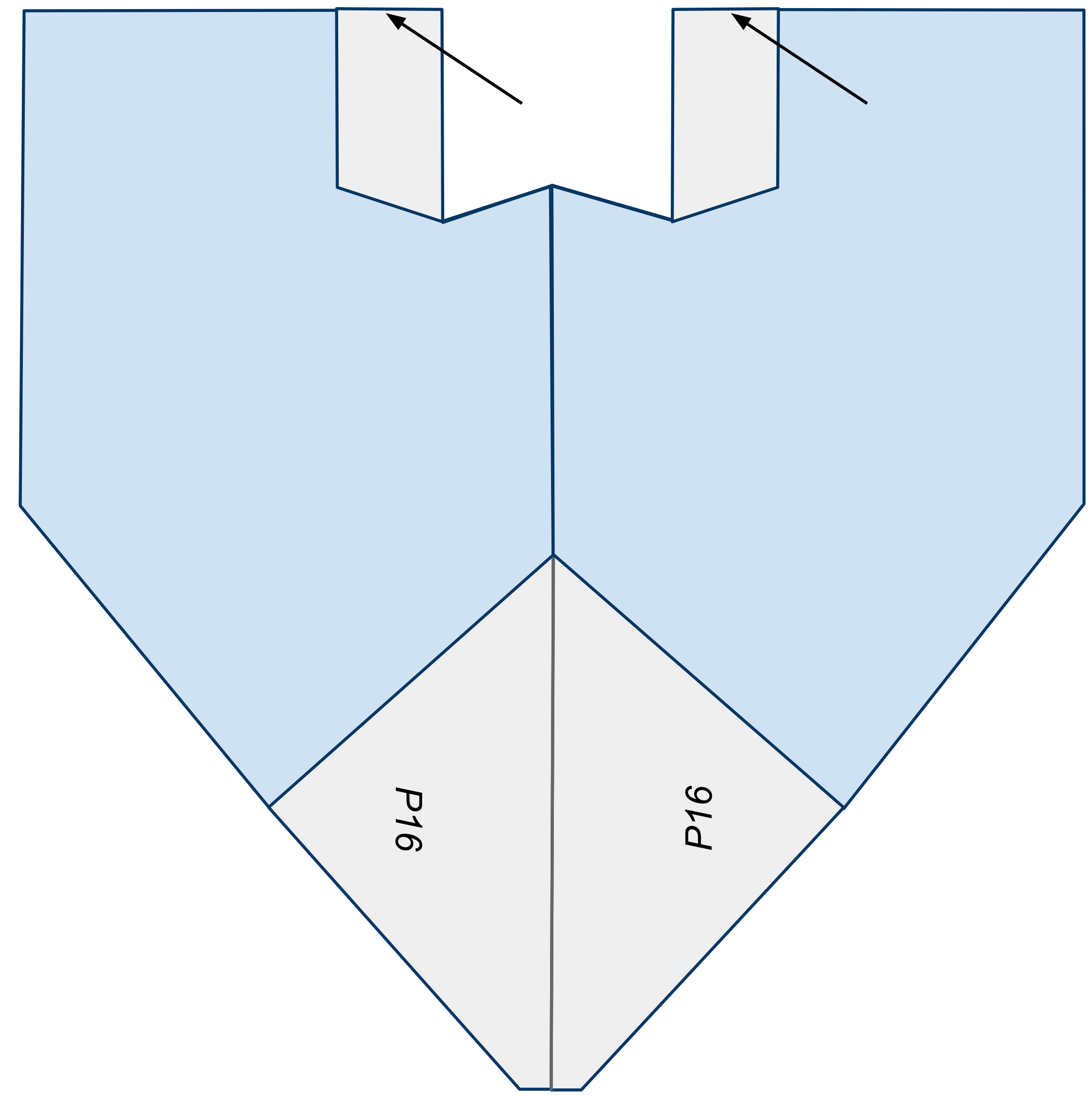
Figure 10



11

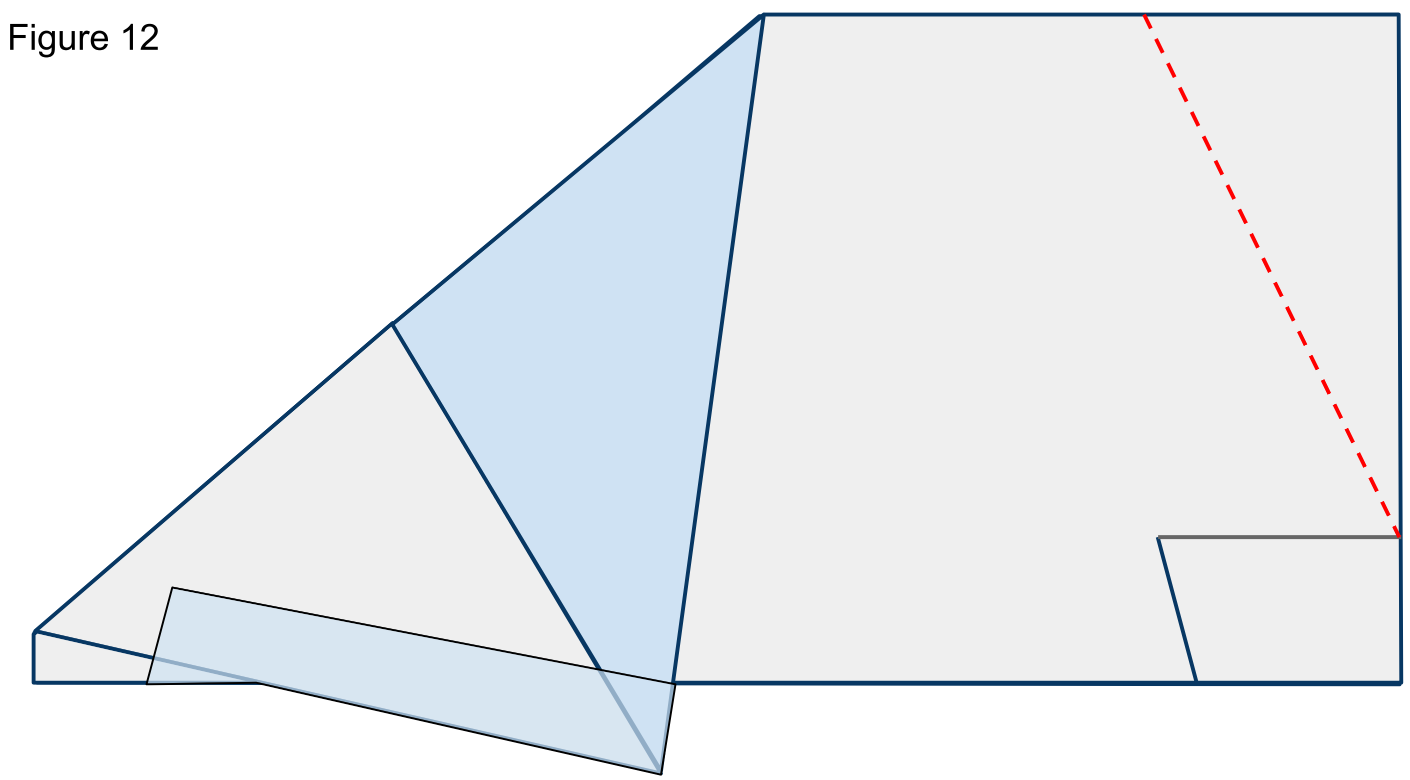
Turn the plane over and fold both tail sections out. Be sure to line up the end of the tail with the end of the wing as shown in Figure 11 below.

Figure 11



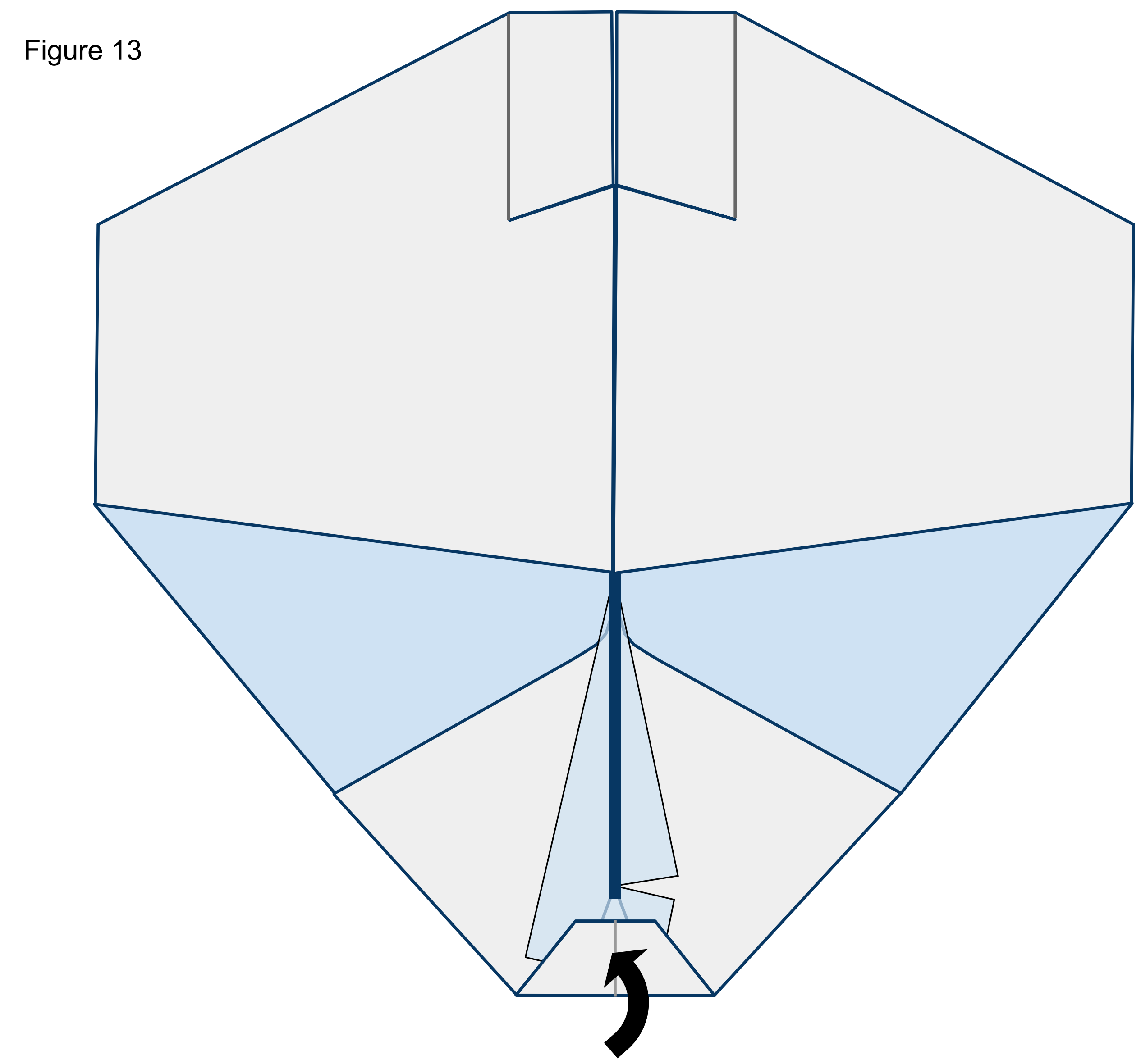
12

Turn the nose to the left and fold the plane back in half. Cut along the red dashed line including both wings as shown in Figure 8 below.



13

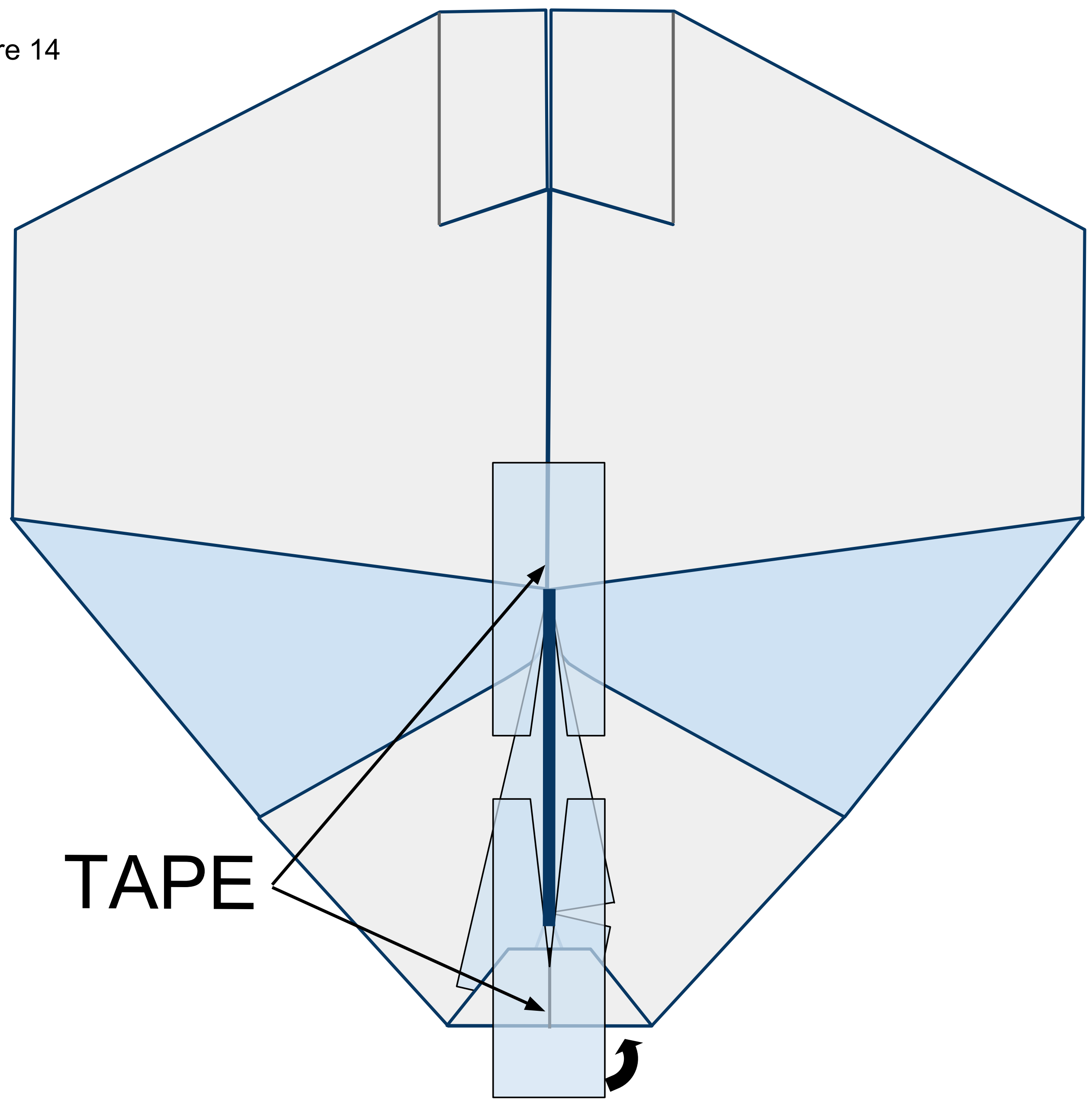
Turn the nose toward you. Open up the wings and lay them flat. Fold the nose of the plane back and in as shown in Figure 13 below. The vertical keel is represented by the thick blue line.



14

Place 2 pieces of tape with slices cut in them on the exposed underside of the plane in front of and behind the keel as shown in Figure 17 below. Wrap the tape around the nose to the top side of the plane.

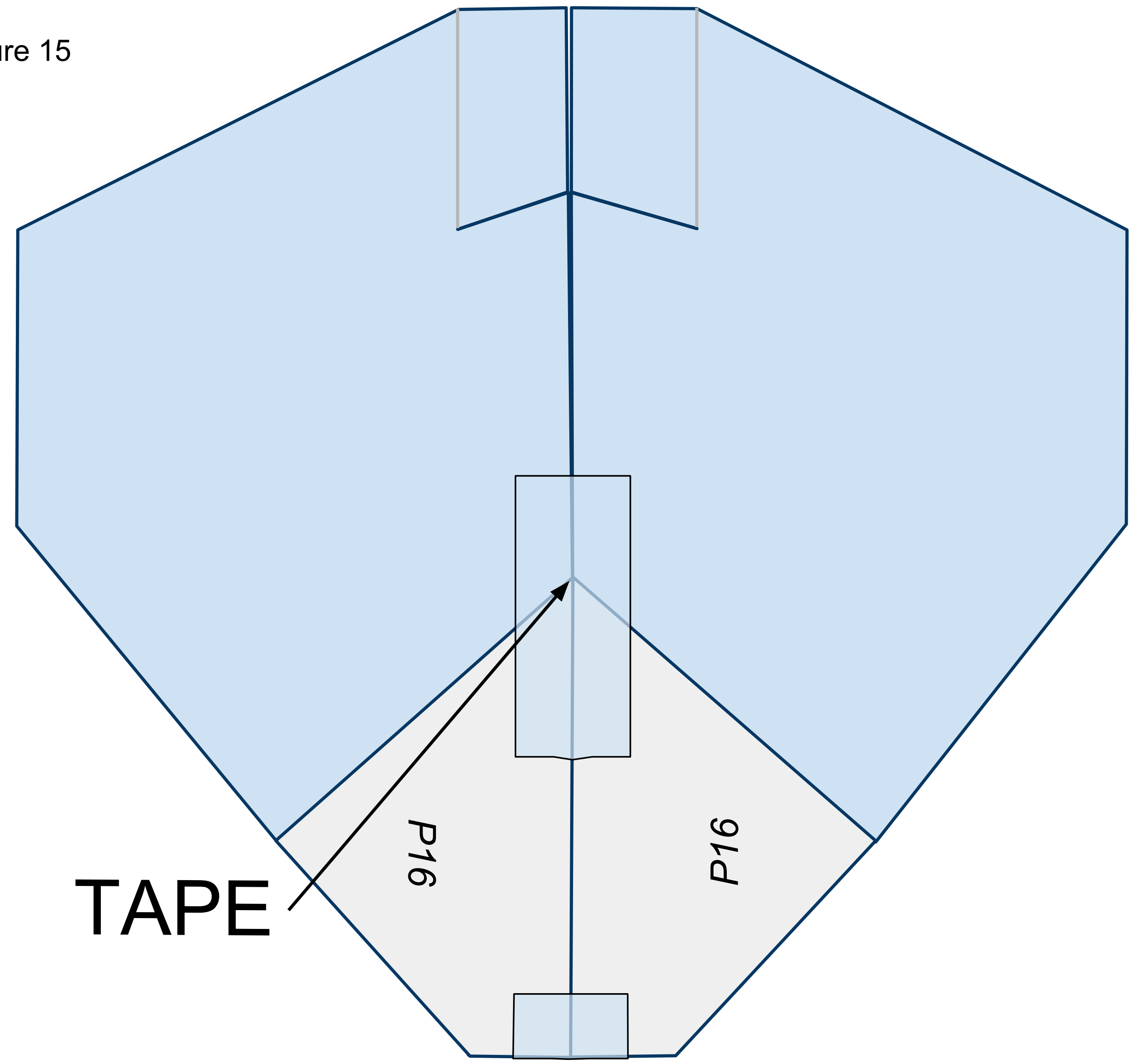
Figure 14



15

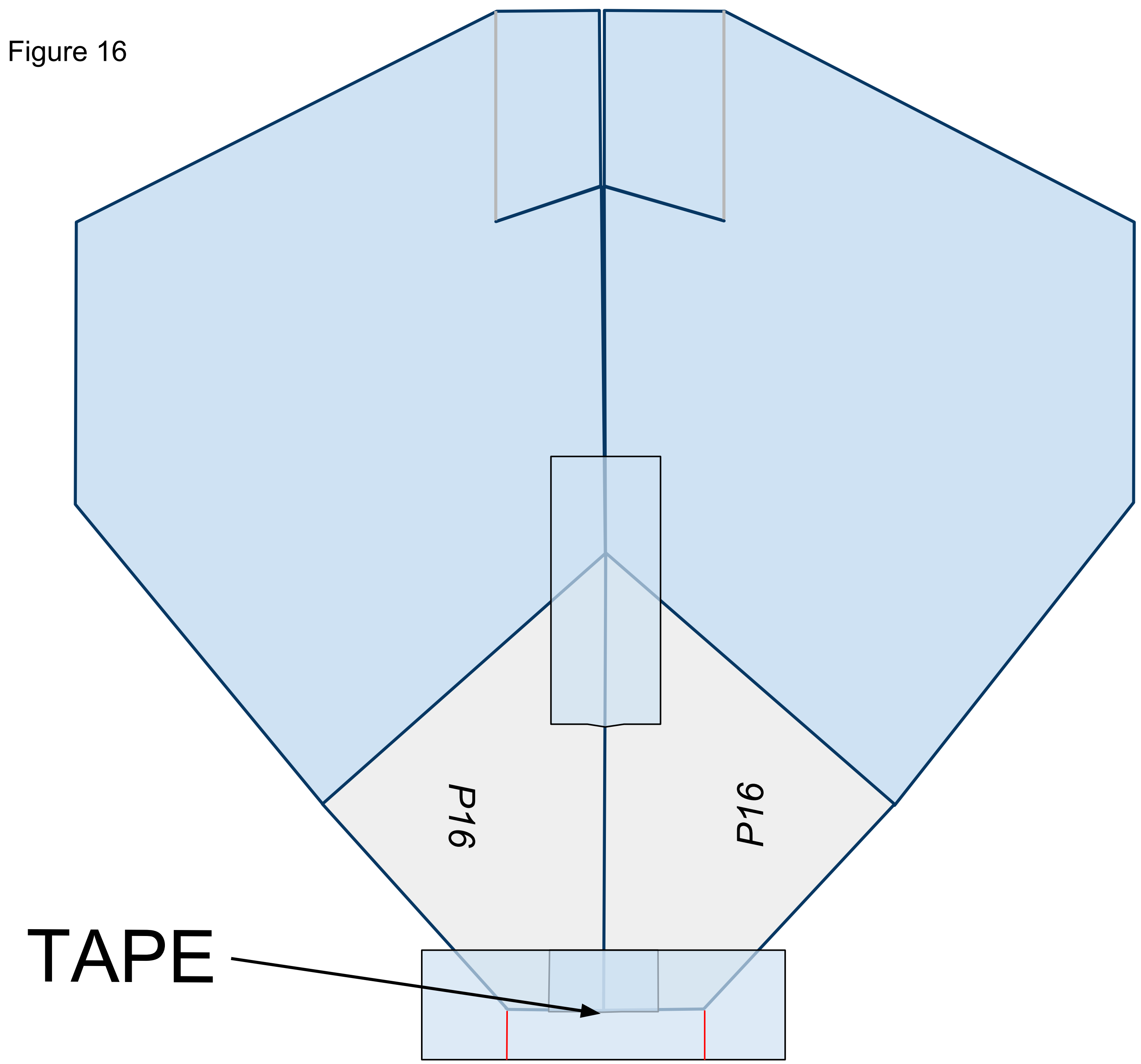
Turn the plane over laying the keel to one side. Place tape along the fuselage as shown in Figure 15 below.

Figure 15



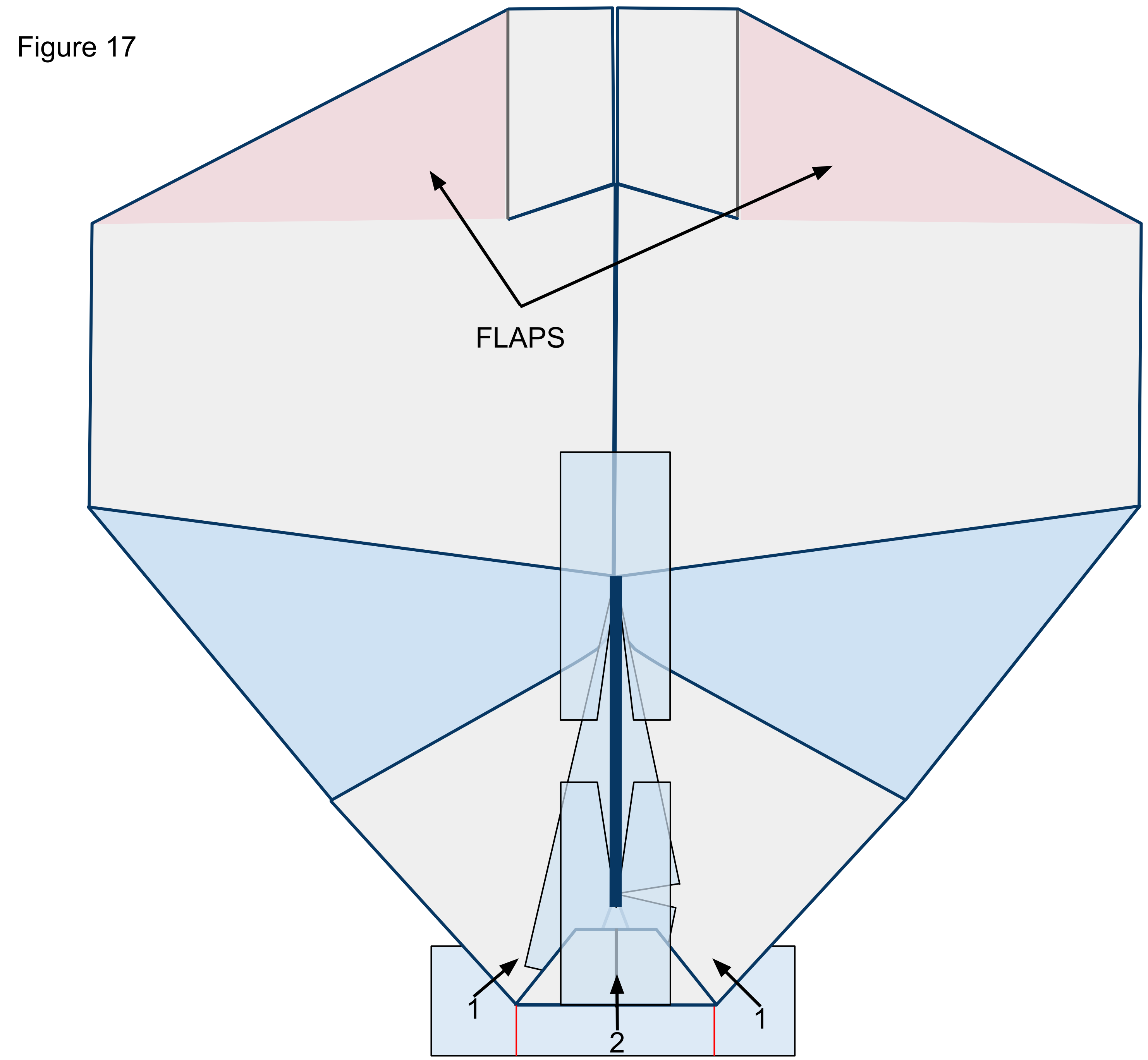
16

Place a piece of tape across the nose of the plane as shown in Figure 16 below. Cut slits along the **red** lines as shown. These slits will help you fold the tape under the plane in the next step.



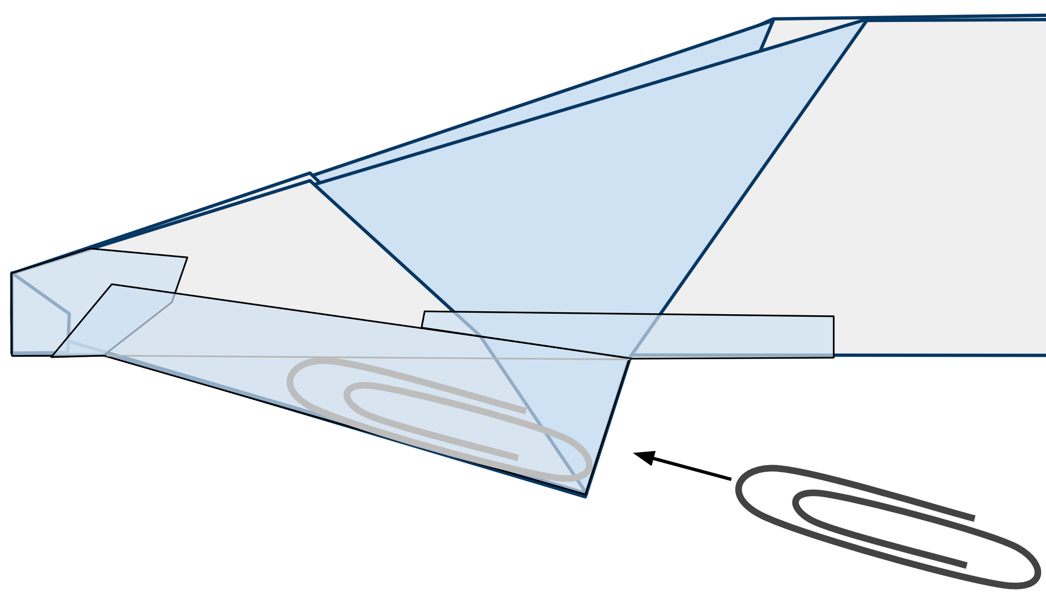
17

Flip the plane over so you can see the top side. Fold the flaps of tape, in number order, up and in as shown in Figure 17 below. They will overlap each other once folded. Also, note the area designated in pink as the flaps for this plane. Don't crease them but simply bend them up or down to control the plane.



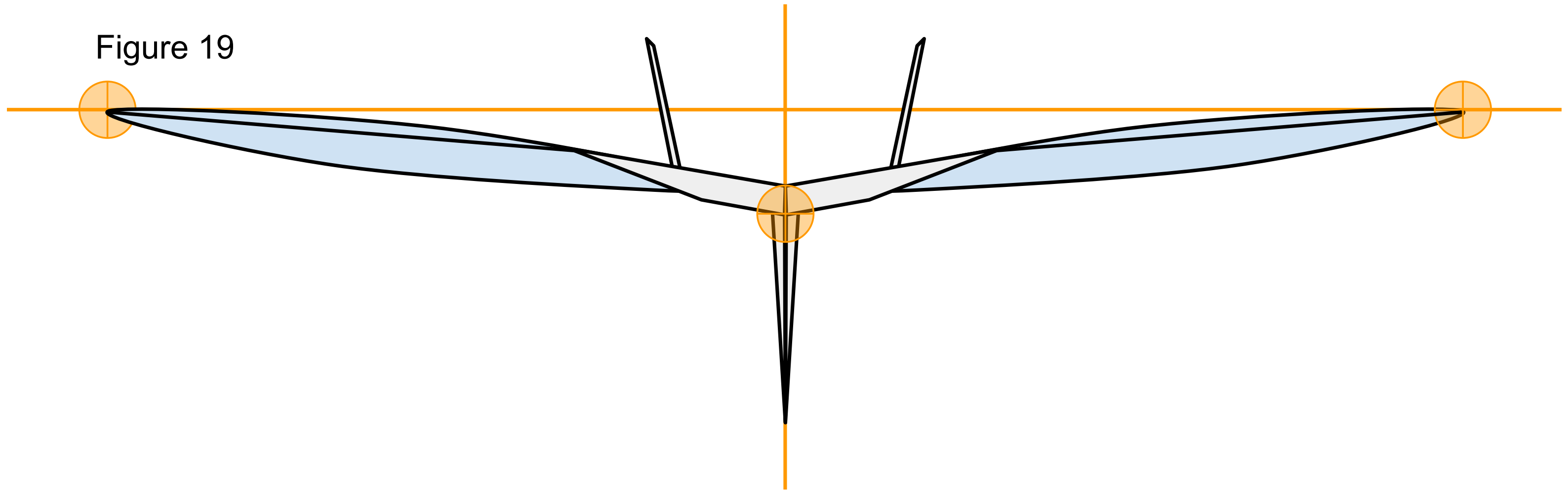
Turn the nose to the left. Insert a #1 size paper clip ***just*** inside the keel through the opening behind it as shown in Figure 21 below. Then seal the opening with a piece of tape.

Figure 18



Now that you have constructed the plane ensure that it has the proper shape. Take a look at your plane head-on from the nose. The plane should be symmetrical. Take a look at Figure 19 below. Note the points of interest. Note the curved shape of the paper on the underside of the wings. To achieve this, carefully stick one side of a pair of scissors up into the pocket on the underside of the wing from behind and gently mold the paper into the proper shape. The pocket under each wing will be stopped at the tape placed across the underside of the fuselage. Don't crease it.

Figure 19



We are almost ready to launch! I recommend taking a quick look at the “PREFLIGHT” section to get some good tips. All paper planes fly erratically on their first flights. They require refining to make them great. To do this, take a look at the “FINE-TUNING” section.